



Recent Advances in Neuropsychological Test Interpretation for Clinical Practice

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Abstract

Much attention in the field of clinical neuropsychology has focused on adapting to the modern healthcare environment by advancing telehealth and promoting technological innovation in assessment. Perhaps as important (but less discussed) are advances in the development and interpretation of normative neuropsychological test data. These techniques can yield improvement in diagnostic decision-making and treatment planning with little additional cost. Brooks and colleagues (Can Psychol 50: 196–209, 2009) eloquently summarized best practices in normative data creation and interpretation, providing a practical overview of norm development, measurement error, the base rates of low scores, and methods for assessing change. Since the publication of this seminal work, there have been several important advances in research on development and interpretation of normative neuropsychological test data, which may be less familiar to the practicing clinician. Specifically, we provide a review of the literature on regression-based normed scores, item response theory, multivariate base rates, summary/factor scores, cognitive intraindividual variability, and measuring change over time. For each topic, we include (1) an overview of the method, (2) a rapid review of the recent literature, (3) a relevant case example, and (4) a discussion of limitations and controversies. Our goal was to provide a primer for use of normative neuropsychological test data in neuropsychological practice.

Keywords Clinical neuropsychology · Reliable change · Norms · Multicultural psychology · Cognition

Background

Recent professional practice surveys show that clinical neuropsychologists have integrated into a variety of medical settings (Russo, 2018; Sweet et al., 2021). With greater involvement in medical systems and interdisciplinary care environments, neuropsychologists have made efforts to adapt

to the modern healthcare landscape. Data suggests that clinical neuropsychologists increasingly employ technologies, such as smartphones and personal computers, to provide assessment and treatment services (Block et al., 2017). Similarly, there was an explosion of scholarship on telehealth practices in response to the coronavirus pandemic, which resulted in new care models (e.g., Hewitt & Loring, 2020) and interorganizational practice recommendations (Postal et al., 2021).

These efforts have deservedly attracted positive attention from the professional community. Less discussed, however, are significant advances that have been made in the development and interpretation of neuropsychological data over the past decade. These improvements should be considered equally critical because they may improve patient care with minimal cost or need for reformatting clinical workflows. Further, as “expert knowledge about the use and appropriateness of different test batteries, inventories, and other methods ... and understanding of their psychometric properties is a core component of the clinical neuropsychologist’s

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toolbox” (Hessen et al., 2018, p. 651–652), routine updates to this knowledge base are critical. Finally, a primary strength of neuropsychologists on interdisciplinary care teams is accurate, objective assessment of cognitive and psychological symptoms (Salvadori & Pantoni, 2020).

Brooks and his collaborators were some of the first researchers to summarize techniques for developing and interpreting normative neuropsychological data and make them easily digestible for practicing clinicians. In particular, their 2009 article, “Developments in Neuropsychological Assessment: Refining Psychometric and Clinical Interpretive Methods,” stands out as a thorough methodological overview and practitioner’s guide. In brief, Brooks et al. (2009) made the following points:

1. To ensure accurate normative comparisons, the sample on which norms are derived must be large and representative of the patient in question.
2. Test score interpretation is influenced by score reliability and distribution shape. Test scores that produce smaller standard errors of measurement and closer approximations of normality yield greater confidence in the accuracy of interpretation for a given score.
3. Low normed scores are common when a battery of neuropsychological tests is administered, even among cognitively healthy individuals. As a result, isolated low scores should not automatically be taken as evidence of impairment.
4. In serial assessment, neuropsychologists are encouraged to use statistical methods that evaluate the significance of change over time. So called “reliable change” methods are useful to determine whether change in observed scores at retest reflects a true change in cognitive performance, as opposed to a practice effect or normal variation.

The principles outlined by Brooks et al. (2009) have become standards for the practice of clinical neuropsychology and are regularly cited in board preparation materials (e.g., Stucky et al., 2020) and consensus guidelines for cognitive assessment (e.g., American Psychological Association Task Force for the Evaluation of Dementia and Age-Related Cognitive Change, 2021). In the decade since this article was published, continued developments in the creation of norms, the calculation of normed scores from test batteries, and interpretation of those scores have continued to emerge. These advances will likely lead to significant improvements in neuropsychology practice. However, it is often difficult for busy practitioners to stay abreast of this literature and integrate updated practices into their established routines. Thus, the goal of this manuscript is to provide an update to Brooks and colleagues’ (2009) work using literature from 2010 through 2022.

In line with Brooks et al. (2009), our review focused on advances in the development of norms, normed score calculation, and interpretation of normed scores. We also extend their work by providing practical clinical guidance for implementation of the reviewed methods using a case example. The sections of the manuscript are organized by topic. For each topic, we provide (1) an overview of the method with relevant historical context, (2) an updated rapid review of the literature supporting the use of the method (Pham et al., 2014), (3) an example of how to apply the method for a hypothetical case, and (4) a discussion of the limitations and controversies surrounding the use of the method.

Method

Search

The topics of norm development, norm score calculation, and norm score interpretation are wide-ranging. Our initial searches using broad topic terms in Web of Science yielded over 80,000 results, an impractical number of articles to review. Consequently, we refined the search to obtain only the most relevant articles using the following Boolean phrase: (Topic=(neuropsychol*)) OR Publication/Source Title=(neuropsychol*) AND Title=(“continuous norm*” OR “quantile regression” OR “robust norm*” OR “regression-based norm*” or “regression based norm*” OR interpretation OR “regression-based change” OR “regression based change” OR “reliable change” OR actuarial OR intraindividual OR bayes* OR “base rate*” OR “item response theory” OR IRT OR “process approach”). Use of the broad search term neuropsychol* for the topic or publication title ensured wide inclusion of articles in the field or neuropsychology. The specific search terms in the title selected for articles focusing on advances in norm development, normed score calculation, and normed score interpretation. These terms were generated from topics covered by Brooks and colleagues (2009), input from the authorship team, and input from the editor and reviewers during the manuscript revision process. Note that we explored adding broad search terms to the title search phrase, such as “norm*.” However, this choice added over 1500 articles, the majority of which did not fit the scope of the manuscript based on an initial review. Thus, we retained the more specific Boolean search phrase.

The date range of the search was January 1, 2010, through April 15, 2022, capturing articles published in the 13 years following Brooks and colleagues’ (2009) work. Results were restricted to English language articles using the “Document Types” and “Languages” filters. This search yielded 390 results. All English language articles citing Brooks et al. (2009) as of April 15, 2022, were also added ($n = 62$).

Fourteen duplicate articles were deleted, resulting in a final group of 438 manuscripts for abstract review. Abstracts were managed in EndNote.

Abstract Screening

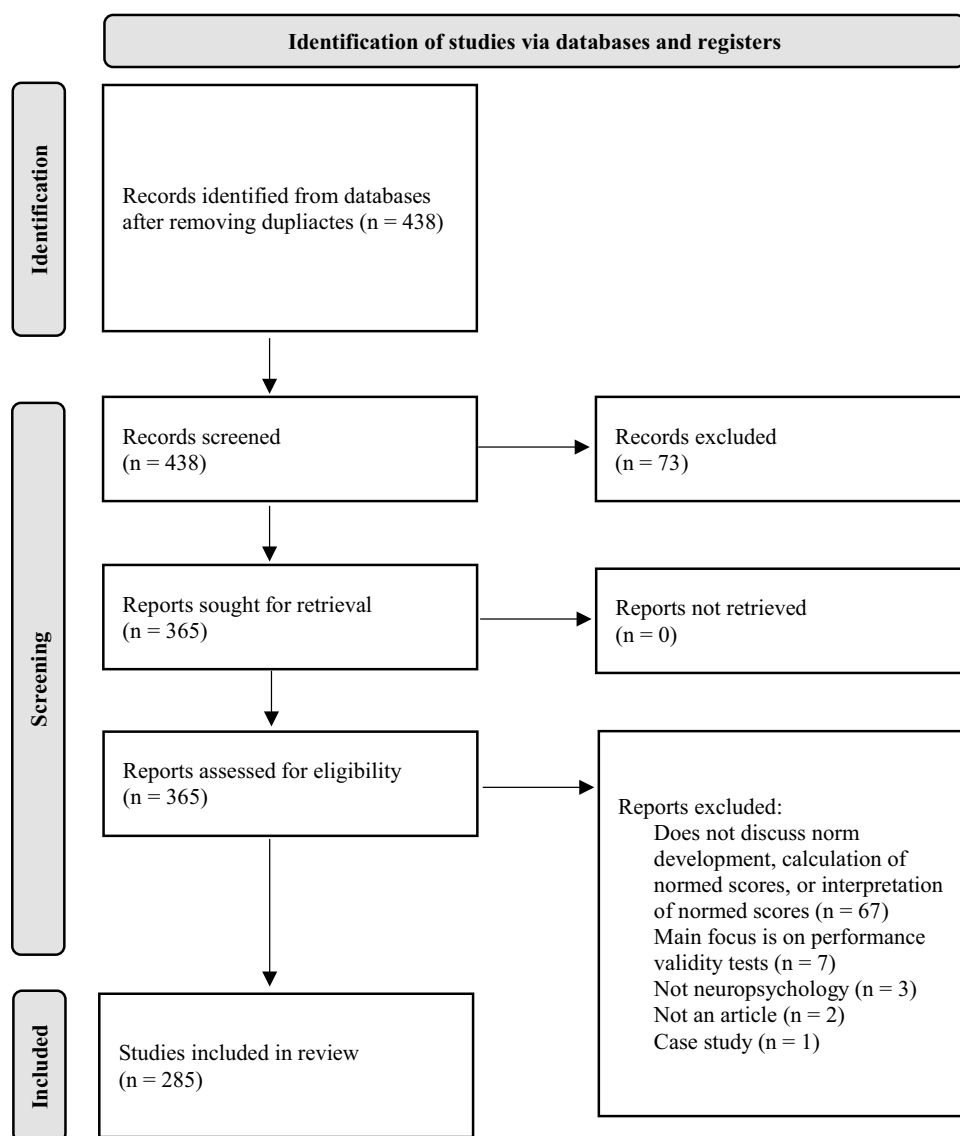
Figure 1 is a PRISMA diagram outlining the results of the article selection process. Abstract screening and full-text screenings were conducted in Covidence independently by at least two of three authors (AK, CM, and RR). To be moved forward to the next stage of the review process, articles needed to meet the following criteria: (1) written in English; (2) include discussion of or data on human participants; (3) include discussion of or data on norm development, calculation of normed scores, or interpretation/use of normed scores; and (4) published in a peer-reviewed journal. Abstracts, books, book chapters, gray literature, case studies,

and studies using single case methods were excluded. Both authors were required to mark an article for removal for it to be excluded. Disagreements were resolved by discussion.

Full-Text Screening

Full-text screening was conducted independently by at least two of three authors (AK, CM, and RR) in Covidence. Both authors were required to mark an article for removal for it to be excluded. Disagreements were resolved by discussion. Reasons for exclusion were documented as follows: (1) source is not in the topic area of neuropsychology; (2) source uses a single case method; (3) source reports on a case study; (4) source does not contain data or discussion on norm development, calculation of normed scores, or interpretation of normed scores; (5) source is not a journal article; (6) source is not written in English; (7) source reports

Fig. 1 PRISMA flowchart of the article selection process



on data only from non-human subjects; and (8) source is not peer reviewed. During the full-text screening process, we encountered several articles that focused on performance validity measures. Because assessment and interpretation of performance validity are its own niche within neuropsychology and have been reviewed by other authors (e.g., McWhirter et al., 2020), we also excluded texts focusing solely on this topic.

Data Extraction

Data extraction was completed by three authors (AK, CM, and RR). A data extraction template was developed with input from all authors and piloted with 10 initial articles to make revisions before finalizing. Once finalized, the data extraction sheet was entered into Covidence and used by all three authors to extract study information. Extractions were completed by two authors for each article, with disagreements resolved at a consensus meeting.

The primary variable of interest for extraction was the type of norm development, norm score calculation method, or interpretation technique used or discussed. First, a determination was made as to whether an article used an advanced technique. If an article simply referred to existing norms or cited norms from a manual, no further extraction steps were taken. There were 17 manuscripts in this category. Articles that used an advanced analysis or interpretation technique were classified into one of the following categories: (1) robust norms, (2) meta-norms, (3) regression-based normed scores, (4) multivariate base rates, (5) item response theory approaches, (6) factor/summary scores, (7) reliable change indices, (8) standardized regression-based change scores, (9) intraindividual variability indices, and (10) other.

A secondary variable of interest was how the analysis or interpretation technique was used in the publication. Categories included the following:

1. Discussion without data. Such articles included any that mentioned an advanced analysis of interpretation tech-

nique but did not analyze data directly. For example, a review paper would fit into this category.

2. Development or demonstration with data. These papers included a demonstration of how a technique might be developed or used in an actual data set with human participants.
3. Validation/application with data. These manuscripts moved beyond a demonstration to provide evidence of the validity or clinical applicability of a technique. Examples included demonstrating that a technique differentiates between known groups, is associated with brain damage, or predicts a clinical outcome.
4. Other. This category was used for manuscripts that did not fit neatly into the three above categories. The most common studies in this category were simulations.

Results of the extraction process are provided in Table 1. Certain categories were collapsed for conciseness and clarity in the table and write-up. First, robust norms and regression-based normed scores were collapsed into the category of norm development/calculation for Table 1, and the review focused on the larger of these topics—regression-based normed scores. And second, reliable change indices and standardized regression-based change scores were collapsed under the heading of evaluating change. Of note, the catch-all “other” category generally included subcategories with a k of 1, making them inappropriate for inclusion as separate topic sections. The only exception wherein there was a large number of manuscripts in a subcategory was for Bayesian approaches ($k = 17$). However, the Bayesian methods were highly heterogeneous, not fitting under a unifying theme. They were therefore not included as a topic section, particularly given that there have been at least two recent reviews of Bayesian approaches to neuropsychology (Huygelier et al., 2021; Parr et al., 2018).

Case Example

As noted above, we refer to a case example within each section to illustrate the use of each neuropsychological analysis

Table 1 Summary of article coding for manuscripts gathered in the initial search

	<i>k</i>	Discussion without data	Development/demonstration	Validation/application	Other
Norm development/calculation	42	1	22	19	0
Item response theory	30	3	8	18	1
Multivariate base rates	67	1	31	33	2
Factor/summary scores	12	1	1	10	0
Intraindividual variability indices	66	1	3	62	0
Measuring change	62	2	18	38	4
Other	31	3	4	22	2

Some articles included more than one assessment interpretation technique and may therefore be counted more than once across categories

or interpretation technique. To increase usability of this manuscript, we use the same hypothetical patient throughout each section. The hypothetical patient is a 70-year-old Black woman with 12 years of education referred by neurology for evaluation of possible dementia.

Throughout the manuscript, we generated and interpreted hypothetical data for this case in line with current guidelines for multicultural neuropsychology practice. As noted by the revised Multicultural Guidelines of the American Psychological Association (APA, 2017), an individual's identity and self-definition are fluid and complex, and this intersectionality is shaped by the social contexts of the individual. Therefore, the ethical practice of neuropsychology must consider the biological, psychological, and sociocultural contexts of the individual across the lifespan. While factors like race and ethnicity are often associated with cognitive performance, such variables only serve as proxies of a complex lived experience (Manly, 2005). From a health disparities' perspective, social determinants of brain health, such as access to and quality of education, socioeconomic status, and literacy, can better provide a window into the underlying mechanisms that help explain disparate performance (Medina et al., 2021; Possin et al., 2021). Moreover, from the perspective of culturally fair practice, neuropsychologists may consider the appropriateness of their tests and norms in light of several cultural characteristics.

Factors pertinent to the fair practice of neuropsychological testing have been summarized under the ECLECTIC framework (Fujii, 2018). Using this approach, a culturally informed evaluator would consider the individual's education (level, quality, and literacy), culture and acculturation level, language spoken (e.g., English proficiency), economic issues (e.g., socioeconomic status and access to care), communication style, testing situation (e.g., the individual's comfort and motivation), concept of Intelligence, and context (e.g., immigration status). Please see Table 2 for an application of the ECLECTIC framework to our hypothetical patient, whom we refer to as Ms. T. Of note, when discussing the case, we describe scores according to consensus guidelines put forth by Guilmette and colleagues (2020).

Topic Section Rapid Reviews

Following extraction, articles were provided to the section authors by topic. Section authors were instructed to provide a rapid review of literature identified in the overall search. However, they were also given freedom to include additional articles that (1) provided historical context on the technique in question, (2) were identified as they reviewed the references and citations of included articles, or (3) were known to them from the literature given their expertise on the topic in question. In each topic section, we provided a rapid review of this literature, selecting articles for inclusion based on

their ability to explain and illustrate the application of the approach in question. Additional references not included in the rapid review narratives are provided by section topic in the supplementary materials.

Regression-Based Normed Scores

Overview

While everyday neuropsychology practitioners may not be involved in the development of norms, knowledge of how normative samples are collected and normed scores are calculated is critical for evidence-based neuropsychology practice (Lezak et al., 2012a, b). In traditional norming approaches, standard scores are calculated based on the normal distribution of scores for a group of cognitively healthy individuals. If demographic variables (e.g., age, education) are thought to influence scores, separate norms may be constructed within each subgroup of interest. From a practical perspective, such approaches can make norming cumbersome because for each variable accounted for in the normative sample, an additional row of cells is required.

For instance, let us imagine a researcher wants to calculate norms across four age bands (e.g., 60–65, 66–70, 71–75, 76–80) separately for men and women. That would mean the researcher would need to calculate eight separate sets of norms (i.e., eight cells in a normative table). As additional variables of interest are added, this number of cells is multiplied by the number of levels for the added variable. For example, if the researcher added an education variable with three levels (e.g., < high school, high school, > high school) to the picture, the number of cells would increase to 24. Given that a minimum number of individuals is needed for each cell, the total sample size also increases dramatically as the number of variables and the number of levels for each variable increases. In the above example, if we assume that a minimum of 50 individuals is needed per cell, the minimum total sample size would increase from 400 (when considering just age and sex) to 1200 (when also considering education). This issue is born out practically in published normative sets. For instance, cell sample sizes for Tombaugh's (2004) normative data for the Trail Making Test range from 13 to 155 when stratified by age and education.

The other issue with the traditional norming approach is that it creates artificial groups from continuous variables. For example, using the age bands mentioned above, a 65-year-old is treated the same as a 60-year-old (a 5-year difference), but different from a 66-year-old (a 1-year difference). Creating such age bands is useful for organization and ease of calculation, but the distinctions between groups must be acknowledged to be arbitrary and often somewhat illogical. Some researchers have attempted to avoid creating such artificial

Table 2 ECLECTIC model for a 70-year-old single, lower middle income Black woman and retired bank teller from the Northeast with a high school education who lives alone and is presenting for neuropsychological evaluation with suspicion of dementia. We refer to her as Ms. T

ECLECTIC model components	Hypothetical case considerations
Education	Ms. T has 12 years of education. She likely has adequate schooling to write, read, memorize, and feel comfortable in test taking situations. However, we recognize that ongoing inequities in educational quality for Black persons mean that education correcting scores may underestimate abilities for this patient if race (or more appropriately, the psychosocial factors associated with race) is not considered alongside education
Culture and acculturation	Ms. T is a 70-year-old Black woman. It should be recognized that most neuropsychological tests were created by White individuals and originally validated in White samples. They often suffer from lack of broader cultural applicability. Therefore, it would be important to be careful to choose culture fair tests where available and make normative comparisons using samples with representative numbers of older Black women
Language	Ms. T grew up speaking and writing in English, such that testing in English would be appropriate. Of course, there are linguistic nuances associated with the Midwest region and the Black community that could impact verbal tests. We would endeavor to adapt testing to accommodate for these differences by, for example, accepting non-standard, but culturally appropriate responses on naming tests
Economics	Ms. T worked for many years as a bank teller and is familiar with computer technology, such that there are not significant concerns that she would be unable to understand task instructions and interpret test stimuli due to socioeconomic factors
Communication	We are aware of research showing that Black older adults are frequently not given adequate time to discuss medical history due to differences in communication styles. It would be important to allow Ms. T time and space to tell the story of her journey to the neuropsychological evaluation
Testing situation	Traditional medical environments are often associated with discrimination and feelings of being dismissed for patients from minoritized groups, leading to a distrust of medical teams. Therefore, we would endeavor to create a welcoming environment for Ms. T (e.g., by having coffee and snacks available, allowing time for rapport building prior to interview) and inviting trusted family members to components of the evaluation. These steps would hopefully build an environment of trust and give Ms. T the best chance to show her true abilities in testing
Intelligence conceptualization	Different cultures do not place equal value on the same characteristics when evaluating intelligence. For example, some cultures view quick responses as portraying intelligence, while other cultures believe intelligent responses are often slow and require reflection. For Ms. T, it would be important to test everyday functional skills and gather collateral information on her performance in the community. These aspects of cognitive performance would likely be more salient to Ms. T than traditional measures of intelligence
Context	Ms. T lives alone and supports herself independently through a retirement account and social security payments. We would be sensitive to the potential economic impacts of the evaluation on her life (e.g., loss of independence) and discuss these with the patient at the outset of the evaluation. We hope she would feel comfortable participating in testing if it is viewed as a way of finding where she stands and developing a financial management plan involving trusted family members

distinctions by creating norms with overlapping cells (e.g., Ivnik et al., 1992). For instance, the norms for a mean age of 65 might be composed of individuals ages 60–70, and the norms for a mean age of 70 might be composed of individuals ages 65–75, leading to some overlap between these two groups.

Methods for calculating regression-based normed scores may address some of the limitations associated with traditional normed scores (Crawford et al., 2012). In regression-based normed score calculation, covariates, such as age and sex, are used as independent variables in a regression to predict raw test scores. Predicted scores are then standardized to create an empirical distribution of scores (i.e., a normal distribution) that are adjusted for the covariates of interest. This process yields demographically adjusted normed scores based on the entire normative sample, instead of a subsample in an individual cell (Shirk et al., 2011). Empirical estimates suggest that traditional normed score calculation requires sample sizes two and a half to five times larger than regression-based

normed score calculation (Oosterhuis et al., 2016). Furthermore, regression-based approaches treat continuous variables as continuous, avoiding the artificial group distinction characteristic of traditional normed score calculation methods. A final advantage of regression-based approaches is that they are more flexible. Many different covariates can be considered in models, and different models can be developed for different populations of interest (Timmerman et al., 2020). Interaction effects among variables can be modeled as well. Finally, non-linear relationships between predictors and test scores can also be captured (Shirk et al., 2011).

Rapid Literature Review

Validity of Regression-Based Normed Scores

Studies that have compared regression-based to traditional normed scores have yielded conflicting results. On the

one hand, several studies reported a reduced likelihood of detecting low scores and classifying individuals with cognitive impairment when regression-based normed scores were used. For example, Berrigan et al. (2014) examined the validity of regression-based normed scores for the oral Symbol Digit Modalities Test (SDMT) in Canadian samples that included 94 control participants, 70 patients with mild MS, and 354 patients with MS of varying severity. The authors reported that regression-based and traditional normed scores demonstrated similar convergent associations with measures of processing speed. However, regression-based normed scores reduced the likelihood of classifying individuals with multiple sclerosis as cognitively impaired. The authors interpreted this finding as a benefit of regression-based normed scores, arguing that they “attenuate[d] the influence of demographic variables on performance” (p. 289). Similar findings were reported by Classon et al. (2021) and Lorentzen et al. (2021).

On the other hand, several studies have found evidence that using regression-based normed scores increases the likelihood of classifying individuals as cognitively impaired. Burggraaff et al. (2017) conducted a comparable study of the SDMT to Berrigan et al. (2014) in a sample of Dutch control participants ($n = 96$) and patients with MS ($n = 157$). They found that using regression-based normed scores yielded a higher likelihood of classifying individuals with multiple sclerosis as cognitively impaired compared with using traditional normed scores. Similar findings have been reported in other samples of MS patients (Parmenter et al., 2010; Smerbeck et al. (2012), healthy older adults (Calderon-Rubio et al., 2021), and patients with brain injuries, mild cognitive impairment, and dementia (Fellows & Schmitter-Edgecombe, 2020).

It is likely that the impact of norm score calculation method on classification of cognitive impairment depends on sample characteristics. For instance, Conti et al. (2015) reported that the cutoff score for the Montreal Cognitive Assessment needed to be adjusted differentially, depending on age and education. Younger individuals and those with higher education required a higher cutoff score for impairment, whereas those with lower education and higher age required higher cutoff scores. Similarly, Stricker et al. (2021) reported that use of regression-based normed scores for the Auditory Verbal Learning Test increased the likelihood of identifying low scores in men but reduced the likelihood of identifying low scores in women.

A number of other studies did not directly compare regression-based to traditional normed scores. Instead, they provided evidence for the validity of regression-based normed scores in isolation. Several focused on the ability of regression-based normed scores to distinguish participants from different clinical groups or cognitive impairment severities. For example, Escudier et al. (2016) developed regression-based norms for the

Judgment Assessment Tool in 30 healthy control participants. They found that normed performance of individuals with mild Alzheimer’s disease ($n = 24$) was significantly worse than performance of healthy controls, with a very large effect size ($d = 1.79$). Regression-based normed scores have also been shown to delineate controls from patients with MS (Marrie et al., 2021) and controls from those diagnosed with ADHD (Malegiannaki et al., 2019). Finally, one study examined the relationship of regression-based normed scores with biomarker indices (Gifford et al., 2018). The authors showed that among 230 healthy older adults, lower regression-based normed scores on the Philadelphia Verbal Learning Test were related to increased neurodegeneration in brain regions involved in memory (hippocampus, entorhinal cortex), as well as with white matter hyperintensity volumes.

Availability of Regression-Based Norms

For tests frequently used in the USA with English speakers, regression-based norms are already in use behind the scenes in computer or web-based scoring programs. It would be difficult to identify every single scoring software that uses regression-based normed scores, and our search focused on primary research articles. However, to provide a few examples, the scoring systems for the Revised Comprehensive Norms for the Halstead-Reitan Battery (Heaton & Psychological Assessment Resources Inc, 2004), the Calibrated Neuropsychological Normative System (Schretlen et al., 2010), and the California Verbal Learning Test, Third Edition (Delis et al., 2017) all use regression-based approaches to generate normed scores. In our review, we also identified a number of non-proprietary formulas or calculators for regression-based normed scores, including for the Brief Test of Adult Cognition (Gurnani et al., 2015), the Auditory Verbal Learning Test (Stricker et al., 2021), the Symbol Digit Modalities Test (Fellows & Schmitter-Edgecombe, 2020), the Repeatable Battery for the Assessment of Neuropsychological Status (Andreotti & Hawkins, 2015; Duff & Ramezani, 2015), the Uniform Data Set 3.0 Neuropsychological Battery (Devora et al., 2019; Weintraub et al., 2018), and the Consortium to Establish a Registry for the Alzheimer’s Disease neuropsychological battery (Lara et al., 2021). Limited regression-based norms for Spanish speakers in the USA are available as well. Please see Morlett Paredes et al. (2021) for a review, as well as the 2022 special issue in *The Clinical Neuropsychologist* on the NP-NUMBRS project. Furthermore, our review demonstrated diversity in the availability of regression-based norms outside the USA (see Table 3).

There have also been some interesting expansions on the use of regression-based techniques for calculating normed scores. For example, some authors have developed methods for combining data across normative samples and using regression to create so called meta-norms (e.g., Demakis, 2006; Mitrushina et al., 2005). Additionally, others have

pioneered regression-based methods for calculating whether scores depart from estimated premorbid functioning (e.g., Asken et al., 2020).

Case Example

Suppose that we are interested in determining whether Ms. T has problems recalling verbal information and administered the Craft Story test (Craft et al., 1996). Examining the raw score, we see she recalled 7 out of 44 details from this story. On its face, this raw score appears low, but we need a normative standard to provide a more objective judgment. Researchers from the Einstein Aging Study provided norms that include a significant proportion of older Black adults, making them potentially appropriate for assessing Ms. T's performance (Wang et al., 2021). Her unadjusted normative performance when compared against this group would place her – 1.56 standard deviations below the mean, corresponding to the below average range. However, the authors provided a calculator for regression-based normed scores that adjusts for sex, age, education, and race. This method places Ms. T's score – 0.82 standard deviations below the mean, corresponding with the low average range. In the first case,

the normative score would likely be cause for concern for problems with recall, whereas in the second case, the score would likely be taken as evidence of relatively preserved recall skills. This latter interpretation only emerges with precise demographic adjustment.

Of course, there are several caveats to this interpretation worth mentioning. First, the interpretation would have to be done with consideration of other factors, especially the referral question and clinical history. Second, we would need to consider the potential impact of the use of adjusted norms on interpretation. The traditional unadjusted normed score may be more appropriate if we are concerned about denying services or compensation, whereas the regression-based adjusted normed score may be more appropriate if we are concerned about over pathologizing. Third, we would acknowledge that the demographic adjustment for education rests on the assumption of homogeneous educational quality, and educational quality can vary considerably depending on sociodemographic background. Finally, we should note that this demographic adjustment could have been accomplished using a traditional normed score approach. However, it likely would have required a much larger sample than the 225 participants included in Wang and colleagues' work.

Table 3 Availability of regression-based norms outside the USA

Study	Tests	Country/region	Population
Argento et al. (2016)	Brief Visuospatial Memory Test-Revised	Italy	Adult patients with multiple sclerosis
Benito-Sanchez et al., (2020a, 2020b)	Rey-Osterrieth Complex Figure; Test de Aprendizaje y Memoria Verbal-Infantil	Latin America and Spain	Community-dwelling Spanish-speaking children
Cavaco et al. (2013a)	Semantic fluency test; phonemic fluency test	Portugal	Community-dwelling adults
Cavaco et al. (2013b)	Trail Making Test	Portugal	Community-dwelling adults
Damasceno et al. (2018)	Brief Repeatable Battery of Neuropsychological Tests	Brazil	Community-dwelling adults
Guardia-Olmos et al. (2015)	10 Spanish-language neuropsychological tests	Latin America	Community-dwelling adults
Inesta et al. (2021)	Raw data for Digit Span; Letters and Numbers; Trail Making Test; Symbol Digit Modalities Test	Spain	Cognitively active older adults
Magnusdottir et al. (2021)	Trail Making Test; Stroop; Verbal Fluency	Iceland	Community-dwelling adults
Neves et al. (2018)	Verbal Selective Reminding Test	Portugal	Community-dwelling adults
Rivera et al. (2021)	Verbal fluency test	Latin America	Community-dwelling children
Toornstra et al. (2020)	Cambridge Neuropsychological Test Battery Delayed Matching to Sample	Ukraine	Community-dwelling children
Van der Elst et al. (2011)	Animal Verbal Fluency (AVF) and Design Fluency (DF) structured and unstructured tests	The Netherlands	Community-dwelling children
Vicente et al. (2021)	Hopkins Verbal Learning Test-Revised; Rey-Osterrieth Complex Figure Test	Portugal	Community-dwelling adults

Discussion

Of course, an appropriate consideration in the use of any norming system is the representativeness of the normative sample in relation to the patient undergoing assessment. Brooks et al. (2009) cover this notion in detail. Furthermore, it should be noted that despite some clear advantages to regression-based normed scores, there is a longstanding debate in the field concerning the use of traditional versus regression-based normed scores (Fastenau, 1998; Heaton et al., 1996a, 1996b; Heaton et al., 1996a, 1996b). For instance, one concern is that a clinician might make unnecessary adjustments. For instance, regression-based approaches often provide corrections for age or education even when these variables do not explain a statistically significant proportion of variance in cognitive performance. Thus, clinicians must carefully consider whether it is appropriate to make no, some, or many adjustments to best characterize the performance of a particular patient on a particular test.

This point was vividly illustrated by recent controversies over race-based norms. The Concussion Injury Litigation case of the National Football League Players brought to light the possibility that Black players were systematically being denied benefits due to the mandated use of race-based norms (“Henry v National Football League,” 2020). The practice of making race-based adjustments to cognitive scores was originally intended to reduce harms from overidentification of cognitive impairment (i.e., false positives). This case brought to light that that false negatives could be equally dangerous (Kiselica et al., 2021a; Possin et al., 2021) by increasing the likelihood of denial of benefits. As such, the utility of race-based norms can vary, depending on the context of their use (Manly, 2005).

A notable weakness of race-based norms is their reliance on a demographic variable with limited scientific value when considered in isolation (Manly, 2005). As a social construct, the relationship between race and brain health is explained by lived experiences, such as discrimination, and related disparities across multiple domains, including personal and systemic factors (Hill et al., 2015). Culturally and linguistically diverse individuals are often excluded from neuropsychological research and are not well represented in the research that develops normative standards (Pugh et al., 2021). Moreover, key demographic variables that help contextualize the lived experience of culturally and linguistically diverse individuals—such as literacy, acculturation, and socioeconomic status—are infrequently reported in neuropsychological literature, despite considerable evidence that these variables better explain differences between ethnoracially diverse groups on cognitive performance (Medina et al., 2021; O’Bryant et al., 2004). Consideration of these and other measurable social determinants of brain health would not only aid in assessing the appropriateness of

applying a set of norms to a specific individual, but it would also aid in the development of better norms (Medina et al., 2021; Possin et al., 2021).

An additional emerging issue for regression-based approaches is the question of what type of regression to use. Standard linear regression is likely best understood by practitioners, but statistical methodologists have argued that neuropsychological test data often violates the assumptions inherent to linear regression, such as normality and homoscedasticity of regression residuals (Sherwood et al., 2016). Semi-parametric (Abbott et al., 2019; Sherwood et al., 2016), non-parametric (Crompvoets et al., 2020; Lenhard et al., 2019; Timmerman et al., 2020), and Bayesian (Wang et al., 2020) approaches to norm score development have thus been put forward. These approaches are mathematically complex and lack sufficient evidence for wide clinical applicability at this time but may come into greater use as they gain further research support.

Finally, debate remains about how to select the most appropriate normative group from which to derive normed scores. For instance, there is an ongoing discussion in the literature of the advantages and disadvantages of conventional versus robust normative groups. In conventional approaches to creating normative groups, cognitive performance is examined for samples of putatively cognitively healthy individuals assessed cross-sectionally (Lezak et al., 2012b). This approach has been criticized because cross-sectional samples are likely to include some individuals who eventually go on to develop cognitive impairment (Ritchie et al., 2007), a particular concern in samples of older adults (Harrington et al., 2017). The presence of such individuals in a normative group creates the possibility that subtle cognitive impairments are being labeled as normal. Consistent with this idea, prior research has shown that individuals who go on to develop MCI and dementia typically show preclinical signs of cognitive decline (Dubois et al., 2016; Kiselica et al., 2020b; Thomas et al., 2020).

Consequently, some researchers advocate for the use of so-called “robust” norms, which exclude individuals who go on to develop cognitive impairment in longitudinal monitoring (De Santi et al., 2008). Use of robust norms is thought to confer several advantages, including reducing false negatives, increasing sensitivity, and allowing for more accurate demographic adjustment (Kramer et al., 2020; Sliwinski et al., 1996). At least three studies have shown that using a robust normative approach could improve the ability to distinguish older adults without cognitive impairment from those with mild cognitive impairment (Clark et al., 2016; Grober et al., 2015; Pedraza et al., 2010). However, recent data suggested that using robust norms may not improve the ability to distinguish mild cognitive impairment from dementia (Kaser et al., 2022).

In summary, regression-based normed scores provide neuropsychologists with the ability to interpret test performance in a highly individualized manner. However, when using these methods, clinicians must be careful about applying corrections appropriately based on the characteristics of the normative sample and the patient. Furthermore, practitioners should consider the context of the evaluation and the impact that a particular score adjustment might have on interpretation, particularly when sensitive issues surrounding race are involved.

Item Response Theory

Overview

Classical test theory (CTT) models relying on traditional factor analysis should be familiar to most neuropsychologist engaged in developing, understanding the structure of, and interpreting scores from measures. CTT models have been criticized for being misapplied, such as when an exploratory factor analysis is used to examine measure structure when a confirmatory factor analysis could be used to test the reproducibility of an already published model (Kline, 2011). Furthermore, CTT approaches may have disadvantages in comparison to item response theory (IRT) approaches. IRT refers to a series of techniques and measurement models that can vary broadly, but basically evaluate relationships between person parameters (i.e., characteristics of the respondent), item parameters (i.e., characteristics of the items), and a mathematical model that unites the two. See Embretson and Reise (2013) for a full review.

IRT models have several attractive features relative to CTT models, mainly driven by the generalizability of item parameters across samples and the ability to evaluate and select optimum items for a number of particular evaluation circumstances (Thomas, 2019). These techniques have been foundational to standardized educational assessments for decades, while their uptake in clinical situations has been slower (Bilder & Reise, 2019; van der Linden & Hambleton, 1997). A promising development in the applicability of IRT methods has included investment by major funders, such as the National Institutes of Health (NIH), in IRT-based cognitive and psychological assessment infrastructure. Additionally, our literature search revealed numerous labs and scientists applying these techniques to existing and newly developed neuropsychological measures.

Rapid Literature Review

IRT at the Core of Developing New Neuropsychological Measures

As a part of the NIH's 2004 neuroscience research blueprint, the decision was made to invest heavily in a modern

cognitive assessment infrastructure that would create a common currency of cognitive measures across usable across studies and populations (Weintraub et al., 2014). This investment led to the development of the NIH Toolbox, a series of cognitive, emotional, sensory, and motor measures developed under modern measurement models and methods. IRT was used not just for the development of questionnaire measures of emotion and behavior but also sat at the core of developing and scoring the Oral Reading and Picture Vocabulary Tests (Gershon et al., 2014; Lai et al., 2011).

The Oral Reading Test asks an individual to read a series of words orally, and each item is marked right or wrong. And in the vocabulary test, individuals hear a word and select from a screen the picture that matches the word on a tablet. Historically, performance on such measures is assessed by starting at a pre-designated floor level. The participant progresses from this floor, working up sequentially to a ceiling level of performance marked by a certain number of consecutively missed words. Scoring is then accomplished by looking up normed scores in a manual or entering the responses into a computer program.

In contrast, the NIH Toolbox version used an IRT-driven computer adaptive testing approach. In this approach, performance on each individual item of a known set of item parameters informed the algorithm for how difficult the next item should be, allowing for a more efficient identification of the participant's language ability level. For instance, an incorrect response on a moderate difficulty item might lead to administration of a low-difficulty item, while a correct response might lead to administration of a high-difficulty item. This method produced reliable scores that correlated strongly with more traditional neuropsychological measures, yet the IRT-based vocabulary test could be completed in 4 min, versus up to 15 min for the traditional measures (Gershon et al., 2014; Weintraub et al., 2014).

In a related vein, the NIH funded the development of the Executive Abilities: Measures and Instruments for Neurobehavioral Evaluation and Research (EXAMINER) battery (Kramer et al., 2014). The EXAMINER battery is a series of computerized and paper or pencil measures that capture executive functioning and social cognition constructions. While many of these measures will be familiar to researchers and clinicians (i.e., verbal fluency measures and flanker style inhibition tasks), an innovative IRT approach was used to create scores for global executive functioning, as well as subscales. This approach allows for psychometrically matched scales to be utilized and for measures to vary across studies and participants, while still yielding psychometrically robust scores (Kramer et al., 2014).

IRT Approaches to Refining Existing Neuropsychological Measures

In addition to developing new measures, IRT has been increasingly applied to existing neuropsychological measures to

optimize their efficiency in administration. For example, due to the focus in IRT on individual item parameters, traditional cognitive assessment items can be evaluated to identify items that have the same difficulty (redundant) and optimize ordering of items for administration, leading to briefer measures. Such an approach has been demonstrated with the Judgment of Line Orientation test (Calamia et al., 2011) where an IRT analysis demonstrated that nearly a third of the items could be omitted. Similar approaches have been taken for the executive interview (Jahn et al., 2015; Midden & Mast, 2017), the progressive matrix test (Chiesi et al., 2012), a structured interview of cognitive problems in schizophrenia (Reise et al., 2011), and a concussion screening tool (Wilmoth et al., 2020). Of note, although several studies suggest that many neuropsychological measures may be too long, Draheim et al. (2018) conversely found that commonly used measures of working memory may have too low of a ceiling.

IRT Approaches to Addressing Assessment Bias

Reducing bias and developing relevant measures for all persons sit as a core challenge of the modern neuropsychological enterprise (Fujii, 2018), and IRT can help reduce bias in assessment through an approach known as differential item functioning (DIF; Holland & Wainer, 2012). This approach rests on the idea that item parameters are different than person parameters. By using all the items on a scale to estimate a person's true ability level, subgroups can be compared item by item to identify items that differentially over- or under-estimate a subgroup's ability level.

Numerous approaches to DIF assessment are possible, though correspondence between approaches tends to be good (Yang et al., 2011). For example, across multiple DIF evaluation approaches, several items on a naming measure were identified that could over- and under- estimate naming ability on a language measure in Spanish versus English speakers (Yang et al., 2011). And, on the Delis-Kaplan Card Sorting Test, considerable DIF was observed by gender for two items (Spencer et al., 2019). Beyond traditional demographics, DIF can also be demonstrated for some items based on socioeconomic variables. For example, items on a children's neuropsychological measure were shown to differ significantly between children in public vs. private school settings (Martins et al., 2020). Null findings from DIF analyses are also important, as they can identify measures that appear to perform equivalently across age, gender, and ethno-racial groups (Cho et al., 2015; Preti et al., 2017). However, it must be acknowledged that a null finding cannot be taken as definite evidence for measurement invariance, necessitating the combination of IRT approaches with Bayesian approaches.

IRT Approaches to Improving the Precision of Neuropsychological Measurement

Unlike CTT scoring approaches that produce values, such as a cumulative count of correct items or total time to complete

a task, IRT approaches to scale and item scoring yield estimates of underlying ability, termed theta (θ). This distinction is important, as not all items give the same estimate of theta or are valid indicators of theta across the spectrum of ability. By focusing on underlying ability estimates, IRT approaches can be used to better understand how neuropsychological measures relate to ability levels and identify means and methods to improve the precision of scores.

Over the past decade, these precision IRT measurement models have been applied in many different populations, especially among patients with dementia or neurodegenerative disease (Balsis et al., 2018; McGrory et al., 2014). Highlighting a few specific findings relevant to neuropsychologists, Balsis and colleagues (2012) demonstrated how traditional scoring on one of the most common cognitive ability measures for Alzheimer's disease clinical trials leads to significant heterogeneity in true score estimates. Similarly, Tsai and colleagues (2012) used an IRT approach to demonstrate that the scores on the Montreal Cognitive Assessment are not equally informative across the range of underlying cognitive dysfunction. For example, especially in milder stages of cognitive decline, relatively small changes in scores can equate to large changes in underlying cognitive ability.

IRT Approaches to Measuring Functioning in the Real World

Cognition and cognitive disorders do not exist in isolation, as cognitive dysfunction has important real-world consequences. Historically, measurement of real-world abilities has occurred somewhat separately from development of cognitive measures. Increasingly over the past decade, IRT has been used to join measures of cognition, biomarkers, and function into cohesive and accurate understandings of disease states. For example, Umeda-Karmeyama and colleagues (2019) used an IRT approach to capture collateral reports of activities of daily living, cognitive functioning, and behavioral symptoms in persons with dementia on three separate factor scores, producing a succinct 14-item measure. Further, IRT models of cognition and functioning were co-developed with brief measures given as a part of epidemiological studies, yielding precision measurements of detecting dementia (GBD, 2016 Dementia Collaborators, 2019). Outside of the dementia population, multiple sclerosis functional disability scores were also submitted to IRT evaluative approaches, yielding valid and reliable short-form self-report estimates of multiple sclerosis-related disability (Chamot et al., 2014). Finally, a recent application of IRT methodology revealed that performance on functional based tasks, such as financial calculations and using a microwave, was differentially sensitive to subtle functional declines, while other activity of daily living measures lacked sensitivity to functional impairment (Lowe & Linck, 2021).

Further Advances in IRT Methodology: Applications to Modern and Historical Questions

IRT approaches are not static, and indeed, new models emerge frequently. For example, IRT has been applied to improve assessment approaches in functional magnetic resonance imaging (fMRI). fMRI studies typically involve the administration of long behavioral stimulus blocks in order to get reliable activation patterns. Thomas and colleagues (2013) addressed the question of whether an IRT approach to understanding the items utilized in such paradigms impact fMRI findings. They found that for commonly utilized working memory measures, item difficulty was low enough for clinically impaired patients to utilize. However, these items appeared to lack difficulty to challenge higher functioning participants at a level needed to elicit optimum cortical activation in the scanner. The authors hypothesized that IRT may be a useful way to build and select stimuli to ensure adequate engagement of cortical networks.

IRT has also been applied to update understanding of long studied concepts. For example, recent work applied IRT to the study of signal detection theory, a decades-old method of assessing recognition memory (Thomas et al., 2018). This study revealed potential for bias in measurement with non-IRT operationalized item parameters. IRT has also been applied to handedness measurement, perhaps one of the most basic behavioral functions with implications for brain-behavior relationships. Busch et al. (2010) used IRT to show that mixed handedness features, as well as particular items on a handedness inventory, incompletely operationalized this foundational skill.

Case Example

As can be seen from the above review, IRT methods can be applied to neuropsychology in many different ways. Here, we focus on a particular clinical application for shortening test administration by application of IRT principles. Calamia et al. (2011) provide an excellent example of how an IRT approach can reduce test administration time. On the traditional Judgment of Line Orientation (JLO) measure, which assesses visuospatial skills, patients are asked to make a series of angular judgments across 30 items. Let us imagine that Ms. T obtained a 22 on this full scale after being administered all items. This score falls above the typical cutoff score of 21 used to suggest evidence of impairment (Strauss et al., 2006). Calamia et al. (2011) used IRT to reorder the items in order of difficulty. They suggested starting with the 16th most difficult item and using basal and ceiling rules of 6 correct (basal) or incorrect (ceiling) responses during administration. Using this method, a clinician would only need to administer 6 items to reach the cutoff score of 21 and establish evidence of intact visuospatial skills.

This approach would reduce the administration time for Ms. T from approximately 20 min to 4 min, decreasing testing burden without sacrificing validity.

Discussion

IRT approaches are increasingly utilized in neuropsychological research and have led to important insights on creating precise, efficient, non-biased, and ecologically valid measures for both emerging and historical clinical questions. Though we remain hopeful that IRT measures will improve our measurement enterprise, several limitations might mitigate the impact of these techniques on the field.

First, from a conceptual standpoint, IRT approaches rest in their most basic forms on strong assumptions of unidimensionality, and neuropsychological measures are frequently multi-dimensional. Nonetheless, multi-dimensional IRT models are increasingly developed and applied and may yield functionally equivalent results to classical test theory applications (Reckase, 2009). Second, we note that neuropsychologists have been relatively reluctant to adopt new tests and measures (Rabin et al., 2016). Thus, IRT designed adaptive short forms and optimized measures may not be employed in clinical and research settings due to the professional inertia in method selection and the desire to avoid deviations from historical data analysis techniques. Finally, we note the pragmatic issue that IRT approaches to scoring and interpretation require an updated psychometric skill set.

Despite these limitations, we feel that the future is bright for IRT in neuropsychological practice. This optimism is driven by calls from leaders in the field that IRT approaches must be central to future test development and training (Bilder & Reise, 2019). In addition, the recent advent of the National Neuropsychology Network (NNN; Loring et al., 2022) will mean a solid infrastructure for collecting item level data for commonly used measures across diverse patient populations. This initiative will allow for applications of IRT methods on the measures clinicians regularly employ. Through a combination of awareness, training, infrastructure, and appreciation for the benefits of IRT, neuropsychological evaluation should continue to evolve to leverage these powerful techniques.

Multivariate Base Rates

Overview

As noted in the “Background” section, Brooks and colleagues (2009) emphasized that low scores are common even in cognitively healthy individuals. For example, when the ten core subtests were interpreted on the Wechsler Adult

Intelligence Test, Fourth Edition (WAIS-IV), 62.8% of the normative sample obtained one or more scores $\leq$ 16th percentile, 42.6% obtained two or more scores $\leq$ 16th percentile, and 29.8% obtained three or more scores $\leq$ 16th percentile (Brooks et al., 2013). These findings demonstrate that healthy adults commonly obtain one or more scores that would be individually interpreted as low average. This conclusion may seem non-intuitive for some clinicians, who might make univariate score interpretations by comparing individual scores to the normal distribution one at a time. However, when two or more scores are interpreted in practice, as is almost always the case, the normative comparison is based on a multivariate distribution, which may be less familiar to some practitioners.

For many commonly used neuropsychological tests (such as the WAIS-IV, the Wechsler Memory Scale, Fourth Edition (WMS-IV), the Delis-Kaplan Executive Function System (D-KEFS), the National Institutes of Health Toolbox Cognition Battery (NIHTB-CB), and the Uniform Data Set (UDS) Neuropsychological Battery), a performance pattern including one low score is observed among the majority of the normative sample, and having a few low scores is observed among a substantial minority of the normative sample (Brooks et al., 2013; Holdnack et al., 2017; Karr et al., 2017, 2018; Kiselica et al., 2020b). In addition to those just listed, multivariate base rates have been prepared for many test batteries used in research and clinical practice, including the Wechsler Intelligence Scale for Children, Fourth Edition (WISC-IV) (Brooks & Iverson, 2012); the Test of Verbal Conceptualization and Fluency; the NEPSY (Developmental NEUROPSYchological Assessment) Second Edition (NEPSY-II) (Brooks et al., 2010); the Consortium to Establish a Registry for Alzheimer's Disease-Neuropsychological Assessment Battery (CERAD-NAB) (Mistridis et al., 2015); the Collaborative European NeuroTrauma Effectiveness Research in Traumatic Brain Injury (CENTER-TBI) Neuropsychological Test Battery (Stenberg et al., 2020a); the Canadian Longitudinal Study on Aging (CLSA) telephone battery (O'Connell et al., 2017); the Immediate Post-Concussion Assessment and Cognitive Testing (ImPACT) battery (Houck et al., 2019; Wallace et al., 2020, 2022); and the Automated Neuropsychological Assessment Metrics (Version 4) Traumatic Brain Injury Military (ANAM4 TBI-MIL) (Ivins et al., 2015).

As a result of this research, there are now several well-defined principles of multivariate base rates for clinicians to consider (Brooks et al., 2013): (a) low scores are common when evaluating a battery of tests, (b) low scores increase in frequency with higher cutoffs for defining a low score (i.e., more low scores will be identified if clinicians use $\leq$ 16th percentile to define a low score versus $\leq$ 5th percentile), (c) the number of low scores increases with the length of the test battery (i.e., more low scores are expected with more tests

administered and interpreted), (d) examinees with lower intelligence tend to obtain more low scores, and (e) the frequency of low scores varies by the demographic characteristics of the examinee (e.g., low scores are more common with fewer years of education).

Rapid Literature Review

In terms of new directions in multivariate base rate research, scholars have begun examining the frequency of low scores in different languages of assessment. Studies have presented multivariate base rates for the Czech version of the UDS Neuropsychological Battery (Nikolai et al., 2022) and a German-language battery called the Materials and Norms for Neuropsychological Diagnostics (Tanner-Eggen et al., 2015). Other studies have focused on Spanish-language neuropsychological assessment among normative samples from the USA (Karr et al., 2021b) and internationally through Latin America and Spain (Benito-Sanchez et al., 2020a, b; Benito-Sanchez et al., 2020a, b; Calderon-Rubio et al., 2021; Inesta et al., 2021; Rivera et al., 2019, 2021).

Another new insight from multivariate base rate research pertains to the *other side of the bell curve*, by assessing the normal frequency of high scores within a normative sample (Karr & Iverson, 2020; Karr et al., 2020, 2022). These studies show that the principles of low score multivariate base rates apply to high scores as well, in that high scores are common, and they vary in frequency based on the cutoff for defining a high score, the number of tests administered and interpreted, and the intelligence and education of an examinee.

The clinical utility of high score base rates may be in the assessment of high-functioning people. Among these individuals, the *absence* of high scores, as opposed to the presence of low scores, may be indicative of cognitive decline. That said, even among high intelligence and high education groups, there is a high prevalence of low scores (Heyanka et al., 2013; Jeffay et al., 2021). Thus, there is nuance in test profiles among high-functioning examinees. For instance, clinicians should keep in mind that a single low score may be common, but a complete absence of (relatively) high scores may be uncommon.

An additional new approach to multivariate base rates has been in the use of flexible algorithms for operationalizing the psychometric criterion for mild cognitive impairment. This approach looks at the base rate of multiple performance patterns (e.g., 4 or more scores $\leq$ 25th percentile; 2 or more scores $\leq$ 5th percentile) and determines the base rate for how often participants in the normative sample meet one or more of these performance patterns. Each algorithm occurs at a frequency approximating the psychometric criterion for mild neurocognitive disorder (i.e., roughly a base rate of < 16%

or ~1 SD below the mean). The algorithms are flexible in that different algorithms can be applied based on estimated premorbid functioning or education, meaning not every examinee would have the same one-size-fits-all criterion for defining cognitive impairment. They also combine low- and high-score base rates, allowing researchers and clinicians to determine whether a test profile has an abnormal presence of low scores or an abnormal absence of high scores.

For example, using the five tests from the English-language NIHTB-CB, an algorithm was designed for participants with high average crystallized ability (i.e., between the 75th and 90th percentiles). Among this portion of the normative sample, 15.5% of participants had a test profile that included one or more of the following performance patterns: 5 scores < 63rd percentile, 4 or more scores < 50th percentile, 3 or more scores ≤ 25th percentile, or 2 or more scores ≤ 16th percentile (Iverson & Karr, 2021). Therefore, if an examinee with high average crystallized ability obtained any one of these performance patterns, they would meet the psychometric criteria for mild neurocognitive disorder according to the algorithm. Similar algorithmic approaches have been put forth to identify preclinical and clinical conditions in age-related neurodegenerative diseases (Jak et al., 2016; Kiselica & Alzheimer's Disease Neuroimaging Initiative, 2021; Kiselica, Kaser, et al., 2021a, b; Kiselica, Webber, et al., 2020b; Oltra-Cucarella et al., 2018; Thomas et al., 2020).

Another new direction in multivariate base rate research involves its application to clinical questionnaire data. Recent studies have evaluated multivariate base rates of low scores on the Behavior Rating Inventory of Executive Function, Second Edition (BRIEF2) and Adult version (BRIEF-A) (Aita et al., 2021, 2022) and the Minnesota Multiphasic Personality Inventory, Second Edition Restructured Form (MMPI-2-RF; Odland et al., 2015). Among the BRIEF2 standardization sample, 37.5–42.2% of healthy children had one or more elevated scores on self-, parent-, and teacher-report ratings of executive functions (Aita et al., 2022). Among healthy adults, 34% obtained one or more elevated scores on the MMPI-2-RF and 26.5–40.7% obtained one or more elevated scores on the BRIEF-A self-report.

Clinically, an appreciation of multivariate base rates can reduce risk of misdiagnosis of cognitive impairment (Nikolai et al., 2022) and prior research has indicated improved diagnostic accuracy of the actuarial approach to diagnosing mild cognitive impairment among clinical populations, such as those with Alzheimer's disease or Parkinson's disease (Fountain-Zaragoza et al., 2021; Graves et al., 2020; Masse et al., 2021; Ophey et al., 2021; Yu et al., 2021). Furthermore, counts of low scores have been used to operationalize cognitive impairment in children and adolescents with lupus (Williams et al., 2011a, b), patients with dystonia undergoing presurgical evaluations (Niccolai et al., 2020), and children in psychiatric inpatient settings (Gaudet et al., 2019).

Case Example

Imagine that Ms. T arrived to the evaluation with her sister, Ms. Z. We may be interested in finding evidence of possible neurobehavioral changes in Ms. T and choose to administer the BRIEF-A informant report to further assess this possibility. Upon examining the results, we find that she had an isolated clinical elevation (i.e., $T \geq 65$) on the emotional control scale. This elevation could suggest clinically significant problems with lability and rash reactions to strong emotions. However, if we examine base rates of clinical elevations for the BRIEF-A standardization sample reported by Aita et al. (2021), more than a quarter of participants have at least one clinically elevated BRIEF-A scale. Thus, Ms. Z's report of her sister's symptoms may be fairly typical, as opposed to a cause for concern. It would be important to gather additional information from Ms. T and Ms. Z about specific examples of emotion regulation problems and their impact on her daily functioning before making further judgments about the clinical meaning of this isolated scale elevation.

Discussion

Despite evidence of clinical utility of base rate approaches, there have been barriers to incorporating multivariate base rates into regular clinical practice. For example, there has been concern that base rates of low scores among healthy standardization samples do not reflect the base rates of low scores expected in clinical practice, where patients present with existing cognitive concerns and have a greater likelihood of obtaining low scores (Gavett, 2015). In line with this criticism, there have been relatively few studies on base rates in clinical settings, although some have been published, as reviewed above.

Another barrier that has limited the utility of multivariate base rates in clinical practice is the perceived requirement of using a fixed neuropsychological test battery for their appropriate use (e.g., Heaton & Psychological Assessment Resources Inc, 2004). That is, there is a belief that if a clinician administers just one more (or less) test than the battery used in preparing the multivariate base rates, then the base rates would no longer be directly applicable to test interpretation. Researchers have attempted to circumvent this issue by simulating the frequency of low scores on a test battery based on published intertest correlations, rather than empirically deriving the frequency of low scores based on observed data (Crawford et al., 2007, 2011; Goette et al., 2023). This approach has minor bias (Decker et al., 2012) and may have clinical utility when working with examinees with average intelligence. However, it underestimates the number of low scores among examinees with low intelligence and overestimates the number of low scores among examinees with high

intelligence (Brooks & Iverson, 2010). Another approach to this problem, which has yet to be explored, may be to report the percentage of low scores, rather than the number of low scores. The percentage of low scores for a population may hold constant regardless of the number of tests administered, though empirical research is needed to confirm this assumption.

Summary Scores

Overview

Summary scores for cognitive functioning attempt to quantify broad cognitive abilities as assessed across several tests. They have been an ongoing focus of assessment research, likely originating with psychometric *g* and the interest in global metrics of overall intellectual ability. The use of summary scores of cognitive performances remains common in neuropsychological practice, with the two most frequently used assessment instruments (Rabin et al., 2016) for adults (i.e., WAIS-IV and WMS-IV) and the most frequently used assessment instrument for children (i.e., the Wechsler Intelligence Scale for Children) providing an overall summary score (i.e., FSIQ) or separate indices for different cognitive dimensions (e.g., Processing Speed Index, Auditory Memory Index). The most commonly administered battery of executive function tests, the D-KEFS, does not provide summary composite scores (Delis et al., 2004). This choice was deliberate by the test developers, based on prior research showing limitations of factor analysis for evaluating construct validity on process-oriented tests (Delis et al., 2003). Nonetheless, researchers have provided some support for latent dimensions of the D-KEFS (Karr et al., 2019; Latzman & Markon, 2010) and prepared composite scores for this measure (Crawford et al., 2011; Iverson et al., 2020a, b). Of note, these composite scores are designed to summarize test data and are not based on any theoretical model of executive functions. Additionally, there have been other important updates in the literature on summary scores, which we turn to now.

Rapid Literature Review

The literature on summary and factor scores is heterogeneous. A number of studies were primarily psychometric in nature. Several explored the underlying factor structure of common intelligence batteries (Benson et al., 2013; Decker et al., 2014; Watkins, 2017). Additionally, one study examined the impact of adding supplemental memory measures to the RBANS Delayed Memory Index (Gradwohl et al., 2022). And Gaudet et al. (2021)

investigated psychometric properties of composite scores from the Immediate Post-concussion Cognitive Test (ImPACT).

Several other studies were more clinically focused. A number of these efforts had the goal of developing and validating composite scores for use in clinical trials of persons with Alzheimer's disease and related dementias. These composite scores were created by analyzing data from batteries of tests administered to large cohorts of older adults. For example, researchers created a Preclinical Alzheimer's Cognitive Composite Score and memory and executive functioning composite scores using data from the Alzheimer's Disease Neuroimaging Initiative (Donohue et al., 2014). Other scholars created factor scores for multiple domains of cognitive functioning in the Uniform Data Set Neuropsychological Battery (Kiselica et al., 2020b; Staffaroni et al., 2021). Beyond older adults, Brydges et al. (2021) used fluid intelligence scores to better understand ADHD symptoms. They found total ADHD symptoms and hyperactive symptoms to be significantly negatively correlated with fluid intelligence, while there was no significant association for inattentive symptoms.

Outside of the clinical context, a paper by Anell and colleagues (2014) had an interesting applied focus on employee selection. They investigated whether a measure of *g* could be derived from a limited paper and pencil test battery using data from 1344 applicants to Swedish police academies. The authors found good fit for a bi-factor model that included a *g* factor and reported that this *g* factor accounted for over 60% of the variance in grade point averages.

Case Example

Given that neuropsychologists commonly use measures with summary index scores, like the Wechsler tests (Rabin et al., 2016), they will likely be familiar with their application in clinical practice. We provide one example of the use of summary scores, which is in the interpretation of scattered test scores within a cognitive domain. Imagine that Ms. T reports significant fatigue and expresses concern about her ability to participate in extended testing. We may then administer a brief test battery, like the RBANS, and interpret the patient's performance on this battery against representative norms for Black older adults (e.g., Andreotti & Hawkins, 2015). In assessing her recall performance on delayed memory subtests, we see that Ms. T's scores are quite variable, ranging from the below average range to the average range (Table 4). We might initially conclude that her recall skills are impaired due to the below average score and the low absolute raw scores. However, we could be thrown off by the fact that two of the three scores are within the broadly average range. Looking up the Delayed Memory Index would reveal a score of 66 (exceptionally low range).

Table 4 Hypothetical Repeatable Battery for the Assessment of Neuropsychological Status (RBANS) delayed memory subtest scores for Ms. T

Subtest	Raw score	Scaled score	Descriptor range
List recall	2	10	Average
Story recall	2	5	Below average
Figure recall	5	7	Low average

After examining this score, our confidence in the interpretation that Ms. T has impaired recall skills may increase, given that this index score summarizes performance across several delayed recall tests.

Discussion

Although summary score methods may be useful to assess overall test performance, they can mask nuance in cognitive test data that neuropsychologists are trained to interpret. For example, the WMS-IV Auditory Memory Index includes logical memory and verbal paired associates, both immediate and delayed recall, in its calculation. By the index score alone, clinicians could not discriminate whether an examinee had difficulty with a given test (e.g., struggled more with story learning than paired word learning) or had difficulty with delayed recall versus immediate recall. Such alternative interpretations of test data may lead to different diagnostic considerations and different recommendations for compensatory cognitive strategies to implement in everyday life. For example, if a patient presents with a relative strength in paired word learning, then using associate learning techniques may be an appropriate compensatory recommendation to assist with daily remembering.

It is also worth noting that there are other approaches for summarizing a test profile that we did not discuss. Such approaches have included an overall test battery mean that averages norm-referenced scores (Miller & Rohling, 2001) and a deficit score that applies weights to low scores and averages those weights to produce an overall score (Carey et al., 2004). Additionally, in the section on multivariate base rates, we described how the overall count of low scores can be used to assess performance across a battery of tests. To date, research on both deficit scores and multivariate base rates has found strong correlations between these scores and overall test battery mean scores (Iverson et al., 2020a, b; Iverson et al., 2020a, b; Stenberg, Karr, Terry, et al., 2020a, b). Thus, different overall summary scores may be redundant without clear advantages of one method over another, though further empirical investigation of this topic is needed.

Cognitive Intraindividual Variability

Overview

Another method of summarizing performance across a neuropsychological battery is to calculate an intraindividual variability (IIV) score. IIV in neuropsychological test scores can arise from a variety of different sources, including characteristics of the tests (e.g., measures with low ceilings), the scores (e.g., varying normative adjustments), and the test taker (e.g., Hill et al., 2013). There is potential interpretive value in observing high levels of cognitive IIV, which is commonly referred to as “dispersion” when used to as a measure of across-test variability (Hilborn et al., 2009).

Dispersion-based measures of IIV have been in existence since the early 1980s (McLean et al., 1990). Dispersion-IIV is usually calculated as the within-person standard deviation of the observed test scores, which may be adjusted for mean level of performance resulting in a dispersion coefficient of variance ($CoV = \text{individual standard deviation} / \text{individual mean}$). See Table 5 for a case example. Although dispersion-IIV values tend to have small-to-medium associations with mean level of performance, they nevertheless are thought to provide incremental diagnostic, prognostic, and ecological value beyond traditional global mean scores (e.g., Morgan et al., 2011). At the conceptual level, higher dispersion-IIV values are hypothesized to reflect variations in maintaining top-down attention control processes (i.e., cognitive dyscontrol) that are regulated by prefrontal systems (e.g., Hilborn et al., 2009). In other words, an individual may have the capacity to execute a particular cognitive task successfully, but may nevertheless show fluctuations in the ability to harness the necessary cognitive resources to do so consistently over time (Stuss et al., 2003).

Rapid Literature Review

Our literature review revealed a growing number of studies that support the construct validity and clinical promise of cognitive dispersion-IIV. For example, there is evidence for the discriminant validity of dispersion-IIV. Recent works show elevations in dispersion-IIV among persons with brain injury, including young adults with sports-related concussions (Rabinowitz & Arnett, 2013), veterans with mild traumatic brain injury (mTBI; Sorg et al., 2021), and persons with more severe TBI (Hill et al., 2013) compared to control participants. Similarly, individuals with neuropsychiatric conditions, such as substance use disorders (Arce Rentería et al., 2020) and major depression (Manning et al., 2021), also show greater dispersion-IIV than comparison samples. Elevations in dispersion-IIV have also recently been observed in infectious diseases that can affect the central

Table 5 Examples of a neuropsychological profile with high IIV-dispersion

Measure	T-scores
CVLT-3 total trials 1–5	30
CVLT-3 delayed free recall	17
RCFT copy	18
RCFT delayed	35
TMT Part A	40
TMT Part B	26
WCST perseverative errors	45
SDMT total	36
COWAT total	37
Animal fluency	44
WMS-IV Digit Span	31
PASAT-50 total	30
BNT total	34
JLO total	26
Grooved Pegboard (dom)	45
Grooved Pegboard (nondom)	42
Mean	29.4
Standard deviation	6.5
Coefficient of variance	0.22

BNT Boston Naming Test, *COWAT* Controlled Oral Word Association Test, *CVLT-3* California Verbal Learning Test Third Edition, *dom* dominant, *IIV* intra-individual variability, *JLO* Judgment of Line Orientation, *nondom* non-dominant, *PASAT-50* 50-item short form of the Paced Auditory Serial Addition Task, *RCFT* Rey-Osterrieth Complex Figure Test, *SDMT* Symbol Digit Modalities Test, *TMT* Trail Making Test, *WCST* Wisconsin Card Sorting Test, *WMS-IV* Wechsler Memory Scale Fourth Edition

The coefficient of variance is a measure of across-test variability (i.e., dispersion) within a battery

nervous system, including human immunodeficiency virus (HIV; Morgan et al., 2011, 2012), hepatitis C virus (HCV; Abbott et al., 2019; Morgan et al., 2012), and West Nile virus (WNV; Sheppard et al., 2018).

However, there are some inconsistent findings regarding dispersion-IIV across the cognitive aging spectrum. A recent meta-analysis showed that higher levels of dispersion-IIV are reliably, but modestly, associated with incident cognitive decline and dementia among typically aging adults (Mumme et al., 2021), though caution is urged in interpreting this study due to a failure to include a risk-of-bias analysis. Similar findings for dispersion-IIV and incident cognitive decline are found among individuals with HIV (Anderson et al., 2018), Parkinson's disease (Jones et al., 2020), and major depression (Manning et al., 2021). Among healthy adults, the historical data showed fairly reliable and robust associations between older age and higher levels of dispersion-IIV (e.g., Hilborn et al., 2009). However, our updated literature review yielded mixed findings for age. One study reported the expected positive relationship between older age and

higher dispersion-IIV (Hilborn et al., 2009), whereas others observed nonsignificant, small effect sizes (De Felice & Holland, 2018) or even negative associations (Buczyłowska & Petermann, 2018). One plausible explanation offered for these unexpected findings was the dedifferentiation hypothesis, which posits that the cognitive profiles of older adults become less heterogeneous over time. This lack of heterogeneity that occurs as performance globally worsens can also help us understand the mixed dispersion-IIV findings in persons with memory disorders. Indeed, some studies show no differences relative to controls (La Plume et al., 2021) and others observe *lower* levels of dispersion-IIV in mild cognitive impairment (MCI) and Alzheimer's disease (AD; Grewal et al., 2021).

The construct validity of dispersion-IIV is also supported by studies showing its relationships with structural neuroimaging and measures of cognition across a variety of populations. Dispersion-IIV is related to microstructural integrity of white matter pathways in older adults (Halliday et al., 2019) and in clinical samples of persons with mild TBI (Sorg et al., 2021) and HIV disease (Jones et al., 2018). With regard to its relationship to measures of cognition, higher dispersion-IIV shows broadly medium associations in the expected directions with global cognition (La Plume et al., 2021; Levine et al., 2018; Rabinowitz & Arnett, 2013; Tanner-Eggen et al., 2015) and information processing speed (Paxton et al., 2021). Higher-order cognitive abilities, such as episodic retrospective memory (Paxton et al., 2021; Thaler et al., 2015), prospective memory (Mustafa et al., 2023; Sullivan et al., 2018), and multi-tasking (Fellows & Schmitter-Edgecombe, 2015), also demonstrate the expected negative associations with dispersion-IIV. Finally, Webber et al. (2022) found that higher dispersion-IIV was associated with greater clinician-rated cognitive fluctuations.

Of greater clinical relevance, dispersion-IIV is associated with a variety of health and everyday functioning outcomes. Higher levels of dispersion-IIV relate to lower overall health status and higher risk of mortality, independent of sociodemographics (Anderson et al., 2018; Thaler et al., 2015). Furthermore, studies in samples with mTBI (Merritt et al., 2022) and West Nile virus (Sheppard et al., 2018) show that greater dispersion-IIV is independently associated with lower health-related quality of life. And several groups reported that suboptimal adherence to antiretroviral medication adherence is associated with greater cognitive dispersion-IIV among persons with HIV disease (Arce Renteria et al., 2020; Morgan et al., 2012; Thaler et al., 2015). Furthermore, higher dispersion-IIV has been associated with worse performance in daily tasks among older adults and individuals with dementia (Fellows & Schmitter-Edgecombe, 2015; Webber et al., 2023). Finally, higher dispersion-IIV is associated with increased likelihood of unemployment (Morgan et al., 2012; Sakamoto et al., 2021).

Case Example

Dispersion measures index variability in performance across an entire battery of tests. To demonstrate their use, one must examine performance across an entire battery; therefore, we generated a hypothetical set of *T* scores for Ms. T reflecting high dispersion (Table 5). In this example, there is significant scatter in performance across tests, with *T* scores ranging from 17 to 45. There is no evidence for global cognitive impairment, based on the average mean *T* score, and we would likely examine the data for patterns. We could, for example, visually assess the scores to see whether there are specific domains of cognition where there is evidence of cognitive impairment (e.g., low scores isolated to the memory domain). Examination of the test data in Table 5 does not show an obvious pattern in this regard. Rather, scores seem to colloquially be “all over the place.” Referencing the elevated coefficient of variance of 0.22, we might characterize the performance as highly variable. If performance validity concerns and obvious other explanations (e.g., sensorimotor impairments, distractions in the test setting, administration or scoring errors) have been ruled out, this variability could be taken as evidence of executive dyscontrol.

Discussion

Taken together, these studies suggest that considering dispersion-IIV as part of a standard clinical interpretation approach may facilitate the detection of some neurocognitive disorders, serve as a cognitive marker of potential problems with real-world functioning, and help to identify persons at risk of incipient cognitive decline. It is important to note, however, that caution has long been raised about interpreting dispersion-IIV as pathological (Watkins, 2003). This criticism centers on reasoning that some degree of variability in test performance is common even among healthy individuals, such that dispersion-IIV may not discriminate between diagnostic groups (Matarazzo, 1990; Silverstein, 1993).

Furthermore, the day-to-day clinical value of dispersion indices is hampered by a few shortcomings in the literature. Perhaps, the most salient issue is that it is difficult to develop norms for IIV. There is considerable variation in the composition of a typical clinical neuropsychological test battery (Rabin et al., 2016), which makes it challenging to generate dispersion norms that would have wide applicability across practitioners. Relatedly, neuropsychologists often use a battery of tests scored from different normative samples, and the impact that these differences have on measures of IIV is unclear. There are also even more fundamental questions about what adjustments might accompany such normative standards. For example, we know less about the influence of key sociodemographic variables that might be factored

into normative sources (e.g., sex, education, ethnicity). Of course, there are exceptions to this critique (e.g., Iverson et al., 2020a, b; Tanner-Eggen et al., 2015), and some test manuals include base rate information on intersubtest scatter (e.g., Wechsler batteries).

Finally, at a practical level, a typical neuropsychological summary sheet will include many different metrics (e.g., *T*-scores, scaled scores, percentiles) that could make it challenging to quickly generate a measure of dispersion. However, as the field moves towards consensus on preferred batteries and reporting practices, dispersion indices may increase in clinical relevance. Furthermore, statistical approaches could emerge that allow for flexible calculation of IIV, depending on the tests administered.

Advances in Assessing Change over Time

Overview

As noted by Brooks and colleagues (2009), clinicians cannot rely on subjective inferences to assess the clinical meaning of changes in test scores over time because change can occur for many reasons, including poor test reliability, practice effects, normal variation, and disease or injury. Therefore, methods for calculating the statistical significance of score changes (i.e., methods of detecting “reliable change”) are of value to practicing neuropsychologists (Heilbronner et al., 2010).

Duff (2012) provided a methodological overview of methods for assessing change over time. He covered the reliable change index (RCI; Jacobson & Truax, 1991), the RCI with an adjustment for practice effects (Chelune et al., 1993; Iverson, 2001), and standardized regression-based approaches (McSweeney et al., 1993). These methods are similar in that they create a standardized frequency distribution of change scores from which statistical inferences can be drawn about the likelihood that a certain level of change is greater than what would be expected due to normal or chance variation. In practice, a clinician would set a prespecified level of change (e.g., $RCI \pm 1.96$), and label change scores that surpass this cutoff as statistically significant. The three methods for assessing change differ in (a) the metric used to standardize the scores, (b) the degree to which practice effects are accounted for in the model, and (c) the extent to which the level of change can be adjusted for other variables, such as demographic factors. Of the three approaches, the standardized-regression based change method is the most mathematically complex but also controls for the greatest number of factors that could influence degree of change. Of note, when using standardized regression-based change scores, many of the caveats about making appropriate adjustments that were discussed in the section on regression-based normed scores also apply.

Updated Literature Review

Since the publication of Brooks et al. (2009), there has been an increase in the availability of reliable change approaches for assessing change over time on neuropsychological measures. Methods for evaluating reliable change have been developed for several individual tests, such as the Boston Naming Test (Sachs et al., 2012), Ruff 2 and 7 Selective Attention Tests (Knight et al., 2010), the King–Devick Test (Elbin et al., 2021), and the Bergen Right–Left Discrimination Test (Grewe et al., 2014). Other studies have put forth reliable change indices for cognitive screeners, including the Montreal Cognitive Assessment (Kopecek et al., 2017), the Mini-Mental State Examination (MMSE; Stein et al., 2012), and the Structured Interview for the Diagnosis of Dementia (SIDAM; Stein, Luppa, Maier, Tebarth, et al., 2012). Furthermore, reliable change approaches have also been applied to whole test batteries, such as the Repeatable Battery for the Assessment of Neuropsychological Status (RBANS; Hammers et al., 2020a, b; O’Connell et al., 2019), the Uniform Data Set Neuropsychological Battery (Gavett et al., 2015; Kiselica et al., 2020a, b, c), and the CENTER-TBI Neuropsychological Test Battery (Stenberg et al., 2020a). Some reliable change methodologies have also been applied to computerized test batteries, including the ANAM4 TBI-MIL (Haran et al., 2016), Cogstate Computerized Battery (Gates et al., 2021), and the Cambridge Neuropsychological Test Automated Battery (CANTAB; Karlsen et al., 2022).

The majority of research reviewed examined reliable change using English-language tests and batteries. However, some studies developed methods for assessing reliable change in different languages, including the Spanish-language MMSE (Fernandez-Calvo et al., 2020), the Spanish-language NIHTB-CB (Karr et al., 2021a), and the German-language Consortium to Establish a Registry for Alzheimer’s Disease (CERAD) neuropsychological battery (Stein et al., 2012).

The abovementioned studies generally provide norms for reliable change indices using cognitively healthy groups, allowing for evaluation of change over time against a controlled standard. However, there is also value in assessing the expected level of change for individuals with brain disease or injury. In this vein, reliable change methods have been applied to various clinical populations. They have included patients with traumatic brain injury (Hinton-Bayre, 2012; Metcalf et al., 2020; Register-Mihalik et al., 2013), amyotrophic lateral sclerosis (Crockford et al., 2018), multiple sclerosis (Barker-Collo & Purdy, 2013; Weinstock et al., 2021), pediatric epilepsy (Busch et al., 2015), partial epilepsy (Nakhutina et al., 2010), Parkinson’s disease (Turner et al., 2016), and subjective cognitive decline, mild cognitive impairment, and Alzheimer’s disease (Rosas et al., 2021).

Research has also validated reliable change indices using biomarkers. For example, in a recent study (Lin et al., 2021),

scores from a regression-based RCI approach (McSweeney et al., 1993) corresponded with cortical atrophy on neuroimaging. Furthermore, reliable change indices have been applied to the study of response to intervention. These investigations have included survivors of pediatric brain tumor (Duda et al., 2020) and breast cancer (Andreotti et al., 2016). In addition, change methods have been used to evaluate cognitive outcomes among patients who have undergone epilepsy surgery (Conradi et al., 2020) or deep brain stimulation implantation to treat Parkinson’s disease (Foki et al., 2018; Mikos et al., 2010; Schoenberg et al., 2012; Williams et al., 2011a, b).

Case Example

As an example of assessing change in clinical practice, suppose that Ms. T presented for testing on two occasions 1 year apart. Let us assume we were interested in language ability, as assessed by the Animals category of semantic fluency (see Table 6). In this hypothetical case, Ms. T’s raw score drops by 4 points over the one-year interval and falls from the low average to the below average range (Weintraub et al., 2018). Our initial impression might be that this level of change provides evidence for a decline in cognition, given the drop in score ranges. However, if we evaluate Ms. T’s change in performance using a standardized regression-based change calculator developed using a representative sample (Kiselica et al., 2020a, b, c), we see that approximately 11% of robustly cognitively normal individuals with a similar demographic background would be expected to demonstrate this level of change in performance over time. Thus, we may be more inclined to conclude that the observed change in category fluency performance represents normal variability within the context of repeated testing.

Discussion

Just as reliable change methods have been examined for many test batteries and clinical populations, there have been many statistical methods developed for detecting reliable

Table 6 Example of cognitive performance over a 1-year interval on category fluency: animal test for Ms. T

Time 1	Raw score	15
	Demographically adjusted %ile rank	20%
	Descriptor	Low average
Time 2	Raw score	11
	Demographically adjusted %ile rank	5%
	Descriptor	Below average

change. There remains debate in the literature about the exact statistical methodology to apply to detect reliable change, without a consensus for the best approach (Hammers & Duff, 2021; Hinton-Bayre, 2010, 2011; Maassen, 2010).

One advancement in assessing change has been in combining the use of well-known reliable change methods with other advanced interpretation approaches. For example, Brooks and colleagues (2016) advocated for evaluation of the multivariate base rates of reliable change scores. They reported that in a sample of robustly cognitively normal older adults, up to 43% of participants had at least one reliably changed score on repeat evaluation using common neuropsychological test batteries (i.e., WAIS-IV; Neuropsychological Assessment Battery). Kiselica et al., (2020a, b, c) reported similar findings using the Uniform Data Set Neuropsychological Battery, and others have replicated these results in pediatric samples (Brooks et al., 2017). Thus, neuropsychologists should be aware of Brooks and colleagues' mantra, "To change is human." In other words, a clinician should expect some number of reliably changed scores in a repeat evaluation and should look for an abnormal number of reliably changed scores to conclude that a clinically significant change in cognitive abilities has occurred.

A final limitation of reliable change scores is that they can only be applied to time intervals reported in research studies. Currently, there are no evidence-based recommendations for test–retest intervals or how to consider practice effects in re-evaluations (Heilbrunner et al., 2010). In our clinical experience, the most common test–retest window is a 1-year time frame, but there is a great deal of variability, and a lack of empirical research on this topic exists. One potential method for dealing with this issue is to retest participants at randomly varying intervals and include number of days since evaluation as a covariate in a standardized regression-based change approach (Kiselica et al., 2020a, b, c).

General Discussion

In the decade since Brooks and colleagues (2009) reviewed the state of the art in normative data development and interpretation, further strides have been made in developing norms and normed scores, summarizing cognitive performance, and evaluating change. These advancements provide neuropsychologists with an evidence-based set of tools to apply when caring for patients. Here, we shift to discussing important practical considerations for using these tools in clinical contexts.

The first point of discussion concerns questions about ecological validity. That is, neuropsychologists using advanced normative methods must distinguish between a particular tool's ability to improve differential diagnosis and prognosis and its ability to predict real-world functioning. The example of driving safety illustrates this lesson clearly.

A fully demographically adjusted regression-based normed score for the Trail Making Test Part B at the 42nd percentile for Ms. T might not be interpreted as evidence of impairment (Wang et al., 2021). However, the raw score corresponding to this percentile would be 200 s. According to the American Geriatric Society guidelines, such a score might be cause for a referral to a driving safety evaluation due to insufficient processing speed and executive functioning capacities needed to safely operate a motor vehicle (Pomidor, 2016). Thus, neuropsychologists must carefully consider the real-world implications of using advanced normative approaches.

Second, neuropsychologists receive years of training to not only consider real-world implications of test findings but also to integrate testing information with data from several other sources (Bieliauskas & Mark, 2017). Due to the advancement of electronic medical records and the proliferation of medical tests and studies, the information available to the clinical neuropsychologist is ever expanding. Practitioners often have access not only to neuropsychological test data and subjective patient and collateral reports but also to laboratory findings, neuroimaging studies, biomarker values, and digital health information. The advanced interpretive methods discussed can be fruitfully combined with other data to draw more firm conclusions. For example, suppose Ms. T has scores falling at the 8th percentile or less on two memory tests. In isolation, these scores may provide evidence of a decline in memory functioning and might be suggestive of Alzheimer's disease. However, if we imagine she also has neurodegeneration observed on brain MRI and hypometabolism in frontal-parietal regions on FDG-PET scan, confidence in the Alzheimer's disease diagnosis might increase (Dubois et al., 2021).

Third, neuropsychologists must apply their understanding of more foundational psychometric concepts to support accurate interpretation of test data. One of the most important of these concepts is that test scores have clinical value insofar as they are a valid representation of cognitive performance. There is widespread recognition that test scores may be invalidated by any number of factors, including poor task engagement, sensory or motor impairment, or fatigue, such that neuropsychologists are encouraged to interpret test data in the context of multiple indicators of performance validity (Lippa, 2018).

More broadly, neuropsychologists would benefit from applying the techniques discussed in this review within the broader framework of evidence-based neuropsychological practice (Bilder, 2011; Chelune, 2010). We are excited to see these efforts play out in the context of continued advances that will arise in the next decade of neuropsychological research. Especially with advances in computer- and smart technology-based testing, the possibility to quantify and validate qualitative aspects of testing increases (Ashendorf et al., 2013; Diaz-Orueta et al., 2020). Regular continuing education to stay updated on these and other developments is thus of paramount importance.

In closing, we must acknowledge certain limitations of this work. First, there are a number of developments in neuropsychology that were beyond the scope of this review. Some of these topics include multivariate normative comparisons (Huizenga et al., 2007), Mahalanobis distances (Tehan et al., 2018), derived and discrepancy scores (Devora et al., 2019), and minimally important clinical differences (Phillips et al., 2015). We are sure other researchers will take up the mantle to cover these approaches in future work. A second limitation of this work was that we completed rapid reviews of several topics relevant to advances in the creation and interpretation of normed scores as an update to Brooks et al. (2009), and the broad scope of this work may have come at the cost of more deeply reviewing the literature on the individual topics. Furthermore, given the sheer volume of literature reviewed, we did not include an evaluation of methodological quality and risk of bias. Thus, further reviews which systematically evaluate the evidence and the quality of evidence for each neuropsychological technique discussed above will be important in future research. Despite these limitations, this manuscript provides clinical neuropsychologist with a key reference for incorporating emerging evidence-based normative techniques into their practices, which will hopefully lead to improvements in patient care.

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Data Availability Not applicable.

Declarations

Ethical Approval This research consisted of a review of the literature. It did not involve human or animal subjects and therefore did not require IRB approval.

Competing Interests The authors declare no conflict of interest.

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